

1 (a)(i)

$$p = \rho gh$$

$$= 1000 \times 9.81 \times 7.0 \times 10^{-2} \text{ or } 1000 \times 9.81 \times 1.9 \times 10^{-2} \quad [C1]$$

$$\Delta p = \rho g \Delta h$$

$$= 1000 \times 9.81 \times (7.0 \times 10^{-2} - 1.9 \times 10^{-2})$$
$$500 \text{ Pa} \quad [A1]$$

1 (a)(ii)

Upthrust is the upward force by the water due to the difference in the pressure between the top and the bottom surface of the object. [B1]

$$U = \Delta p \times A$$

$$U = 500 \times (5.1 \times 10^{-2})^2 \quad [M1]$$

$$= 1.3 \text{ N} \quad [A0]$$

1 (a)(iii)

$$F_n = 0$$

$$T + U = W$$

$$T + 1.3 = 4.0 \quad [C1]$$

$$T = 2.7 \text{ N} \quad [A1]$$

1 (a)(iv)

$$T = kx$$

$$2.7 = 30 \times x$$

$$x = 0.090 \text{ m} \quad [C1]$$

$$\text{height above surface} = 0.090 - 0.070$$

$$= 0.020 \text{ m} = 2.0 \text{ cm} \quad [A1]$$

1 (b)(i)

Newton's second law of motion state that the rate of change of momentum of an object is directly proportional to the resultant force acting on it [B1] and the change in momentum occurs in the direction of the resultant force [B1].

1 (b)(ii)

$$F_n = ma$$

$$4.0 - 1.3 = \frac{4.0}{9.81} \times a \quad [C1]$$

$$a = 6.6 \text{ m s}^{-2} \quad [A1]$$

1 (b)(iii)

As the cube accelerates, its velocity increases and hence the viscous force acting on the cube (by water) increases (and then becomes constant when reach terminal velocity) [M1]. With weight and upthrust being constant, net force and acceleration decreases (to zero) [A1].

2 (a)